

Assignment-3

(using if else, nested if else, for and while loop, continue and break)

1. Write a program to find the **smallest and largest digit** in a number.
2. Write a program to find the **frequency of each digit** in a number.
3. Write a program to check whether a number is **binary or not**.
4. Write a program to count **how many digits are even and odd** in a number.
5. Write a program to print the **multiplication table** of a given number.
6. Write a program to find the **smallest and largest digit** in a number.
7. Write a Python program to construct the following pattern, using a nested for loop.

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8. Print numbers from 1 to N, but **skip numbers whose digit sum is divisible by 3**.
9. Count numbers between 1 and 100 whose **reverse is greater than the original number**.
10. Print all numbers between 1 and 100 where **product of digits > sum of digits**.
11. Print all numbers from 1 to 100 that are **not divisible by any digit they contain**.
12. Print all pairs (i, j) from 1 to 10 such that:
 - i + j is even
 - i * j is divisible by 3
13. Swap the **first and last digit** of a number.
14. Count how many numbers between 1 and 100 contain **exactly two digits same**.
15. Check whether a number is a **Harshad number** (divisible by sum of digits).
16. Find numbers whose **square ends with the same digits as the number itself**. (Hit: $5^2=25$)
17. Find numbers between 1 and 1000 that become **prime after removing one digit**.
18. Print all numbers from 1 to 500 that are divisible by **every non-zero digit** they contain.
19. Print numbers from 1 to 100 but:
 - Skip multiples of 4
 - Stop completely if a number divisible by 17 appears
20. Find the LCM and GCD of two numbers.